Magíster en Estadística, PUCV

Regression with Regularization: Ridge and Lasso

Alejandro Veloz

Overview

1. Least squares and why it fails.
2. Ridge regression: definition, solution, the SVD picture.
3. Ridge as MAP estimation with a Gaussian prior.
4. The lasso: definition and geometry.
5. Why the lasso selects variables: soft thresholding.
6. Computing the lasso: coordinate descent and the path.
7. L_q penalties and the elastic net.
8. Choosing λ in practice.

Notation for this lecture

We use the ESL convention throughout:

$$\mathbf{X} \in \mathbb{R}^{N \times p}, \qquad \mathbf{y} \in \mathbb{R}^{N}, \qquad \beta \in \mathbb{R}^{p}$$

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Careful: in ESL Chapter 3, t denotes the **budget** in $\sum_j |\beta_j| \leq t$ — it is *not* a target. Bishop's targets t_n are ESL's y_i.

Least squares: the starting point

$$\widehat{\beta}^{\mathsf{ls}} = \operatorname*{arg\,min}_{\beta} \ \mathsf{RSS}(\beta) = \operatorname*{arg\,min}_{\beta} \ (\mathrm{y} - \mathrm{X}\beta)^{\mathsf{T}}(\mathrm{y} - \mathrm{X}\beta)$$

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Setting the gradient to zero gives the **normal equations** $\mathrm{X}^{\mathsf{T}}\mathrm{X}\beta = \mathrm{X}^{\mathsf{T}}\mathrm{y}$, so

$$\widehat{\beta}^{\mathsf{ls}} = (\mathrm{X}^{\mathsf{T}}\mathrm{X})^{-1}\mathrm{X}^{\mathsf{T}}\mathrm{y}, \qquad \mathsf{Cov}(\widehat{\beta}^{\mathsf{ls}}) = \sigma^2(\mathrm{X}^{\mathsf{T}}\mathrm{X})^{-1}$$

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By the **Gauss–Markov theorem**, $\widehat{\beta}^{\mathsf{ls}}$ has the smallest variance among all *unbiased* linear estimators.

Why least squares fails

1. **Collinearity.** If columns of X are nearly dependent, $\mathrm{X}^\intercal\mathrm{X}$ is ill-conditioned and Cov$(\widehat{\beta}) = \sigma^2(\mathrm{X}^\intercal\mathrm{X})^{-1}$ has huge entries. Coefficients become large, unstable and of erratic sign.
2. $p > N$. Then $\mathrm{X}^\intercal\mathrm{X}$ is singular: infinitely many solutions, all with zero training error.
3. **Interpretation.** With many predictors we want a *sparse* model we can read and explain.

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All three are fixed by adding a penalty on the size of β.

Ridge regression

Two equivalent formulations (ESL 3.41 and 3.42).

Penalized form

$$\widehat{\beta}^{\text{ridge}} = \underset{\beta}{\text{arg min}} \left\{ \sum_{i=1}^{N} \left(y_i - \sum_{j=1}^{p} x_{ij}\beta_j \right)^2 + \lambda \sum_{j=1}^{p} \beta_j^2 \right\}$$

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One-to-one correspondence between $\lambda \geq 0$ and $t \geq 0$: large $\lambda \leftrightarrow$ small t.

The ridge solution

In matrix form,

$$\text{RSS}(\lambda) = (\mathrm{y} - \mathrm{X}\beta)^\mathsf{T}(\mathrm{y} - \mathrm{X}\beta) + \lambda\beta^\mathsf{T}\beta$$

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which is quadratic in β, so the minimizer is in closed form (ESL 3.44):

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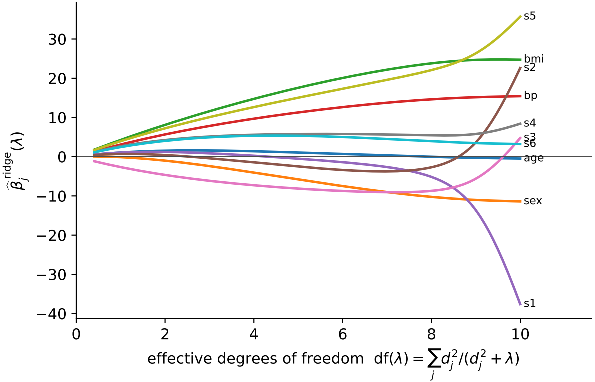
- $\mathbf{X}^\mathsf{T}\mathbf{X} + \lambda\mathbf{I}$ is **always** invertible for $\lambda > 0$, even when $p > N$.
- The problem is strictly convex $\Rightarrow$ a unique solution.
- $\lambda \to 0$ recovers least squares; $\lambda \to \infty$ drives $\widehat{\beta} \to 0$.

Ridge coefficient paths



Diabetes data ($N = 442, p = 10$). Each curve is one $\widehat{\beta}_j$ as λ varies.

Ridge through the SVD

Let $\mathrm{X} = \mathrm{U}\mathrm{D}\mathrm{V}^\mathsf{T}$ with $\mathrm{D} = \mathrm{diag}(d_1 \geq \cdots \geq d_p)$. Then (ESL 3.47):

$$\mathrm{X}\widehat{\beta}^{\mathsf{ls}} = \sum_{j=1}^{p} \mathrm{u}_j \, \mathrm{u}_j^\mathsf{T} \mathrm{y} \qquad\qquad \mathrm{X}\widehat{\beta}^{\mathsf{ridge}} = \sum_{j=1}^{p} \mathrm{u}_j \, \frac{d_j^2}{d_j^2 + \lambda} \, \mathrm{u}_j^\mathsf{T} \mathrm{y}$$

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Ridge projects y onto the principal directions and **shrinks each coordinate by** $d_j^2/(d_j^2 + \lambda) \leq 1$.

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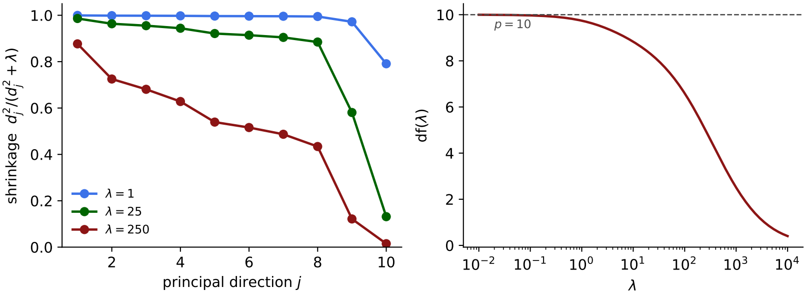
Directions with small d_j — those in which the data has least variance — are shrunk most.

Effective degrees of freedom



$$\mathsf{df}(\lambda) = \mathsf{tr}\left[\mathrm{X}(\mathrm{X}^\mathsf{T}\mathrm{X} + \lambda\mathrm{I})^{-1}\mathrm{X}^\mathsf{T}\right] = \sum_{j=1}^{p} \frac{d_j^2}{d_j^2 + \lambda}$$

Ridge as MAP estimation

Bishop §3.3. Gaussian likelihood, Gaussian prior on the parameters:

$$p(\mathrm{t} \mid \mathrm{w}) = \prod_{n=1}^{N} \mathcal{N}\left(t_n \mid \mathrm{w}^{\mathsf{T}} \boldsymbol{\phi}(\mathrm{x}_n), \beta^{-1}\right), \qquad p(\mathrm{w}) = \mathcal{N}\left(\mathrm{w} \mid 0, \alpha^{-1} \mathrm{I}\right)$$

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$$\frac{\beta}{2} \sum_{n=1}^{N} \{t_n - \mathrm{w}^\intercal \boldsymbol{\phi}(\mathrm{x}_n)\}^2 + \frac{\alpha}{2} \mathrm{w}^\intercal \mathrm{w}$$

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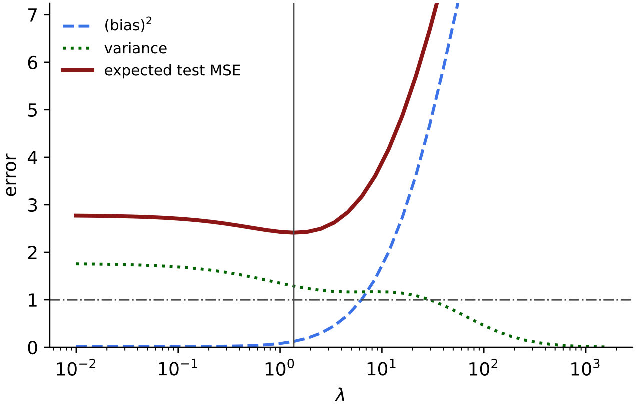
This is ridge regression with $\lambda = \alpha/\beta$.

Ridge and the bias–variance trade-off



Simulation: $N = 40, p = 25$. Bias grows with λ, variance falls, and the sum has an interior minimum.

The lasso

Tibshirani (1996); ESL §3.4.2. Replace the L_2 penalty by an L_1 penalty.

$$\widehat{\beta}^{\text{lasso}} = \operatorname*{arg\,min}_{\beta} \left\{ \tfrac{1}{2} \sum_{i=1}^{N} \left(y_i - \sum_{j=1}^{p} x_{ij}\beta_j \right)^2 + \lambda \sum_{j=1}^{p} |\beta_j| \right\}$$

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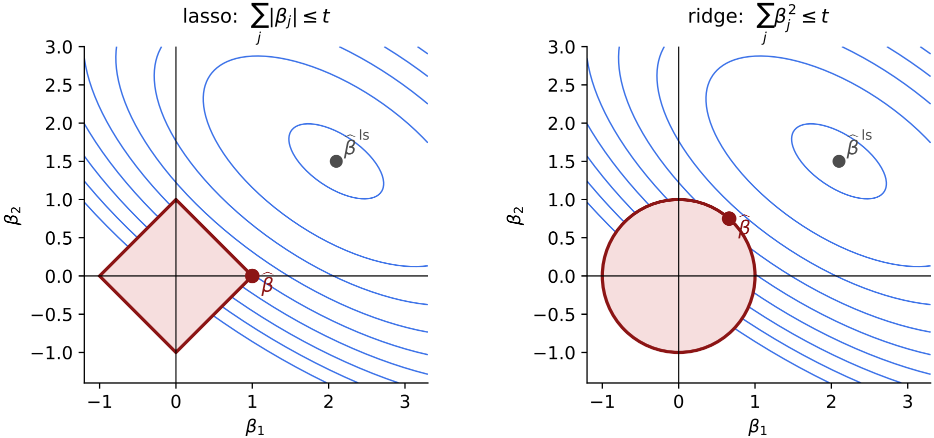
- Still **convex**, so there are no local minima.
- **Not differentiable** at $\beta_j = 0$: no closed form, and that is precisely where the interesting behavior comes from.
- Produces **exactly zero** coefficients $\Rightarrow$ continuous variable selection.

Why the lasso selects: the geometry



Blue: contours of RSS(β), centered at $\widehat{\beta}^{\,\text{ls}}$. Red: the feasible region.

The orthonormal case: closed forms

When $\mathrm{X}^{\mathsf{T}}\mathrm{X} = \mathrm{I}$, all three estimators are explicit functions of $\widehat{\beta}_j^{\text{ls}}$ (ESL Table 3.4):

Estimator	$\widehat{\beta}_j$
Best subset (size M)	$\widehat{\beta}_j^{\text{ls}} \cdot \mathbb{1}(\lvert\widehat{\beta}_j^{\text{ls}}\rvert \geq \lvert\widehat{\beta}_{(M)}^{\text{ls}}\rvert)$
Ridge	$\widehat{\beta}_j^{\text{ls}}/(1+\lambda)$
Lasso	$\text{sign}(\widehat{\beta}_j^{\text{ls}})\left(\lvert\widehat{\beta}_j^{\text{ls}}\rvert - \lambda\right)_+$

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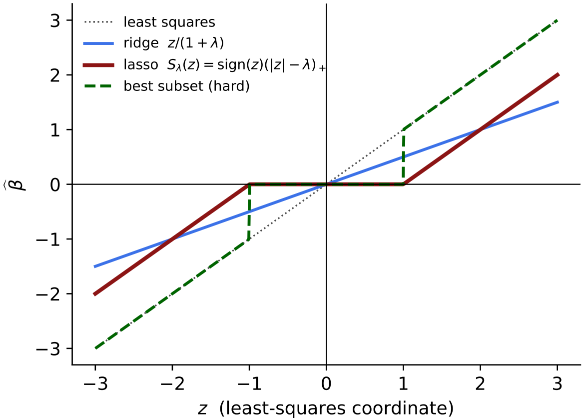
The lasso applies **soft thresholding** $S_\lambda(z) = \text{sign}(z)(|z| - \lambda)_+$; best subset applies **hard thresholding**; ridge applies **proportional shrinkage**.

Thresholding functions



Computing the lasso: coordinate descent

Cycle over $j = 1, \dots, p$, holding the others fixed. The partial residual is

$$r_i^{(j)} = y_i - \sum_{k \neq j} x_{ik} \widehat{\beta}_k$$

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and the one-dimensional problem has the soft-threshold solution

$$\widehat{\beta}_j \leftarrow \frac{S_\lambda\left(\sum_{i=1}^{N} x_{ij} r_i^{(j)}\right)}{\sum_{i=1}^{N} x_{ij}^2}$$

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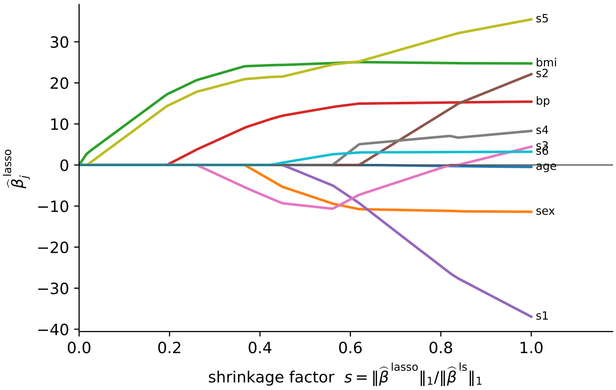
Repeat to convergence. Extremely fast with **warm starts** along a grid of λ (ESL §3.8.6; `glmnet`).

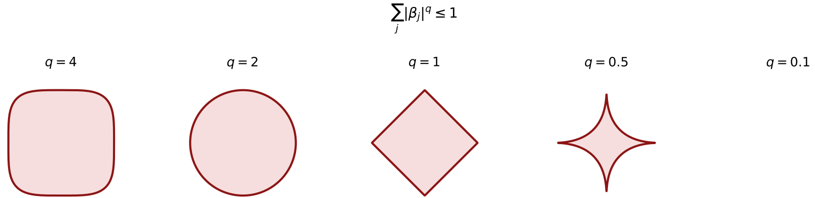
Lasso coefficient paths



Same data as before. $s = \|\widehat{\beta}^{\text{lasso}}\|_1 / \|\widehat{\beta}^{\text{ls}}\|_1$; the path is piecewise linear in λ (LARS, ESL §3.4.4).

L_q penalties: a family

$$\min_{\beta} \sum_{i=1}^{N} \left(y_i - \sum_j x_{ij}\beta_j \right)^2 + \lambda \sum_{j=1}^{p} |\beta_j|^q$$



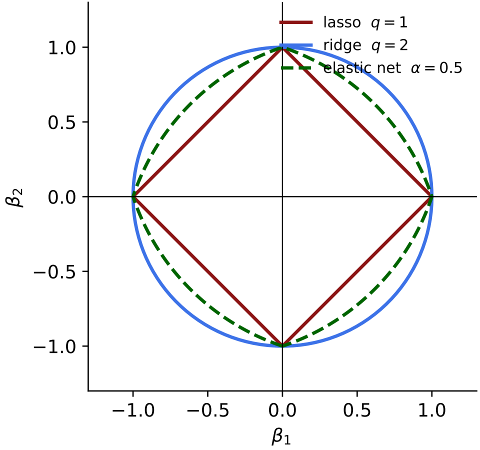
$q = 2$: ridge. $q = 1$: lasso. $q = 0$: subset selection. $q < 1$ is sparser still but **non-convex**.

The elastic net

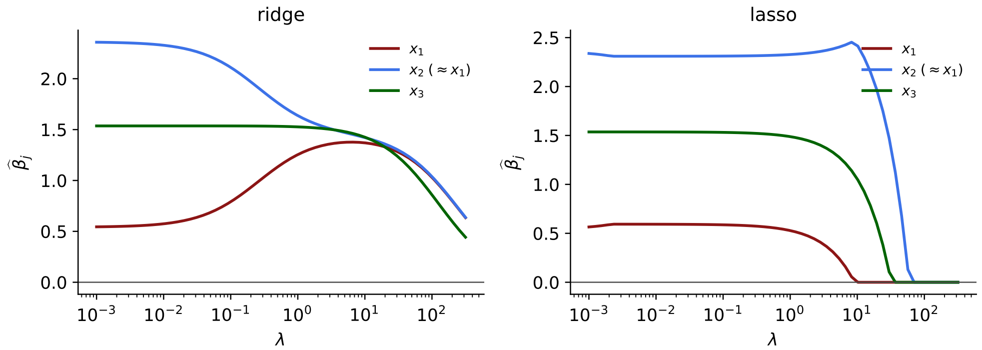
$$\lambda \sum_{j=1}^{p} \left(\alpha \beta_j^2 + (1-\alpha)|\beta_j| \right)$$

Zou & Hastie (2005); ESL §3.4.

- Selects like the lasso, shrinks correlated groups together like ridge.
- Convex; has corners; the boundary is strictly convex between them.
- Handles $p \gg N$ where the lasso saturates at N variables.

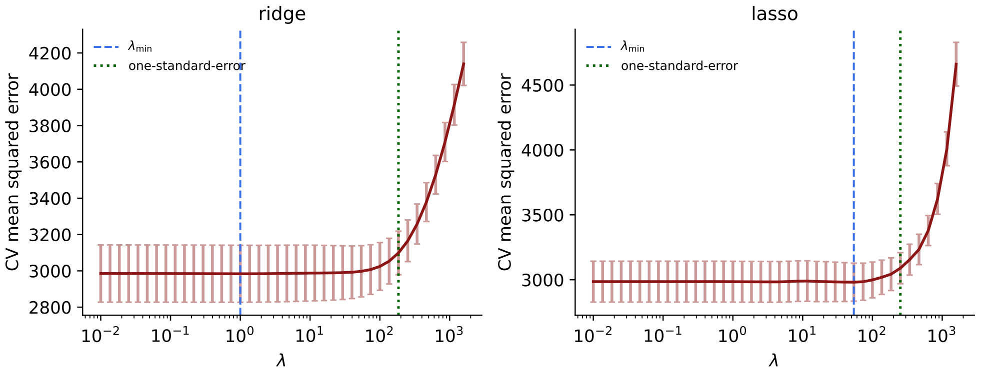


Correlated predictors: ridge versus lasso



x_1 and x_2 are almost identical copies of the same signal.

Choosing λ



10-fold CV. Blue: λ_{min}. Green: the largest λ within one standard error of the minimum.

Ridge versus lasso: summary

	Ridge (L_2)	**Lasso** (L_1)
Penalty	$\lambda \sum_j \beta_j^2$	$\lambda \sum_j \lvert\beta_j\rvert$
Solution	closed form	no closed form (coordinate descent)
Sparsity	never exactly zero	exact zeros; selects variables
Path in λ	smooth	piecewise linear
Correlated inputs	shares weight	picks one, arbitrarily
$p > N$	fine	selects at most N variables
Bayesian prior	Gaussian	Laplace
Best when	many small effects	few large effects

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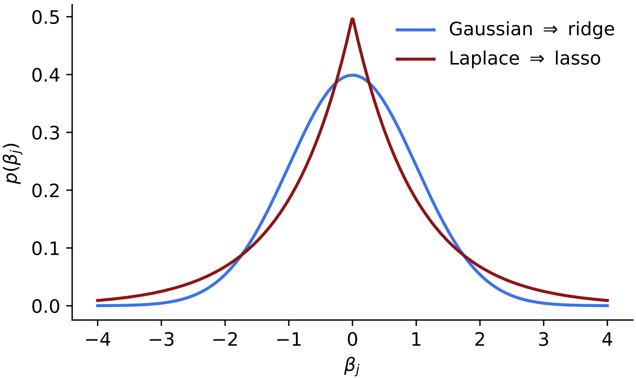
In doubt: try both, plus the elastic net, and let cross-validation decide.

Priors behind the penalties



Ridge $=$ MAP under $\beta_j \sim \mathcal{N}(0, \tau^2)$. Lasso $=$ MAP under $\beta_j \sim$ Laplace$(0, b)$.

Summary

- Least squares is minimum-variance among **unbiased** estimators — accepting bias can do better.
- **Ridge**: L_2 penalty, closed form, shrinks along principal directions by $d_j^2/(d_j^2+\lambda)$; never sparse.
- **Lasso**: L_1 penalty, soft thresholding, exact zeros; $q=1$ is the only exponent that is both convex and sparsity-inducing.
- Both are MAP estimates: Gaussian prior for ridge, Laplace for the lasso.
- Standardize the inputs, never penalize the intercept, choose λ by cross-validation with the one-standard-error rule.